<Voice Coder AI>

(Project Proposal)

Project Code

VCAI560

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Abstract

Traditional programming requires extensive keyboard use, creating barriers for individuals with physical disabilities or repetitive strain injuries who cannot type efficiently [1], [4]. VoiceCoder AI is a voice-controlled web application that converts spoken commands into valid C++ code, implements real-time compilation, provides AI-powered code generation and explanation, and offers accessible programming capabilities without keyboard dependency [1], [2].

Implementation Steps:

- Design web interface with integrated Monaco code editor [3]
- Implement voice recognition using Web Speech API with Noise Gate filtering [1]
- Develop C++ specific voice command parser [5]
- Integrate GPT-4o AI for intelligent code generation and error fixing [12]
- Integrate cloud compilation services [2], [8]
- Implement JWT authentication and PostgreSQL cloud database [7]
- Add Urdu Text-to-Speech support
- Conduct usability testing [4]

Expected Benefits: Provides inclusive programming education tools, reduces physical barriers in software development, and offers a hands-free coding alternative for all programmers, with Urdu language support for Pakistani students [4].

Background and Justification

Existing programming environments such as VS Code and CodeBlocks rely entirely on keyboard input [3]. Some voice programming tools exist such as Serenade and Speech2Code, but they require paid subscriptions, complex installations, or lack AI integration [10], [11].

VoiceCoder AI is a free web-based solution using browser technologies [1], [2]. Unlike existing tools, it integrates GPT-4o allowing users to speak natural language requests like "write a bubble sort function" or "fix errors in my code" and receive working C++ code instantly [12].

Additional enhancements include Urdu Text-to-Speech, cloud file management with shareable links [13], AI autocomplete, push-to-talk mode, and custom voice commands.

Justification: This project removes barriers in programming by providing a zero-cost voice-controlled coding platform [4]. It focuses on accessibility, education, AI integration, and ease of use for students [4], [5].

Project Methodology

Phase 1 — Interface Design
Web interface using HTML, CSS, JavaScript with Monaco Editor [3]. Includes dark theme, syntax highlighting, and multi-tab support.

Phase 2 — Speech Recognition Integration
Web Speech API implemented with continuous listening mode [1]. Noise filtering applied using Web Audio API.

Phase 3 — Command Parsing and AI Code Generation
Three-level voice command system:

Level 1: Direct C++ snippets (loops, classes, print) [5]
Level 2: System commands (compile, save, share)
Level 3: Natural language processed via GPT-4o [12]

Phase 4 — Compilation and Backend
Node.js + Express backend handling API requests. Compiler integration using Judge0, Paiza.io, JDoodle [2], [8]. Server-side proxy protects API keys [7].

Phase 5 — Authentication and Cloud Storage
JWT authentication with bcrypt hashing [7]. PostgreSQL database stores users, code files, and chat history [13]. Users can save and retrieve files via cloud.

Phase 6 — AI Features
GPT-4o enables code generation, debugging, and review [12]. AI autocomplete provides suggestions after inactivity.

Phase 7 — Urdu TTS and Accessibility
Urdu Text-to-Speech using Speech Synthesis API [1]. Push-to-talk mode and custom voice commands included.

Phase 8 — Testing and Bug Fixes
Testing includes voice accuracy, compiler integration, authentication flow, and cross-browser compatibility [4].

Project Scope

In Scope:

- Voice-to-code conversion for C++ [1]
- GPT-4o AI code generation and debugging [12]
- Cloud compiler integration [2], [8]
- Monaco Editor integration [3]
- JWT authentication [7]
- PostgreSQL cloud database [13]
- Unique public share links for code files
- Urdu Text-to-Speech support
- Web-based responsive application

Out of Scope:

- Support for languages other than C++
- Offline voice recognition
- Desktop application for Windows or Mac

High Level Project Plan

Milestones:

1. Setup and Monaco Editor integration [3]

2. Voice capture and basic commands [1], [5]

3. Advanced command parser [5]

4. AI integration (GPT-4o) [12]

5. Compiler backend integration [2], [8]

6. Authentication and cloud storage [7], [13]

7. Urdu TTS and accessibility features [1]

8. Testing and optimization [4]

9. Documentation and presentation

Time Allocation:

Frontend Development: 25%

Voice Processing: 18%

AI Integration: 18%

Backend + Database: 18%

Compilation Integration: 10%

Testing & Debugging: 7%

Documentation: 4%

Hardware & Environment:

Modern browser (Chrome/Edge)

Railway cloud hosting [14]

Supabase database [13]

Success Metrics:

Voice-to-code working for C++ constructs

AI generates correct code from natural language [12]

Compile/run working end-to-end [2]

Authentication and file saving working [7]

Urdu TTS working correctly [1]

High voice recognition accuracy

Final Product:

Live web application deployed on cloud [14]

Full voice-controlled coding system

AI-powered coding assistant [12]

Cloud file management system [13]

Urdu language support

Complete documentation and demo

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